Fpt University

TRƯỜNG ĐẠI HỌC FPT

Requirement & Design Specification

PhucNguyen Resort & Tour Hub

Version: 1.0

– Hanoi, June 2026 –

Record of Changes

Version	Date	A* M, D	In charge	Change Description
V1.0	15/07	A		
V1.0	16/7	M		
V1.0	16/7	A		
V1.0	17/7	A		
V1.0	18/7	A		

*A - Added M - Modified D - Deleted

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I. Overview

1. System Context

PhucNguyen Resort & Tour Hub is a web-based integrated resort management system that combines hotel booking, restaurant services, tour booking, billing, and operational management into a single platform. The system enables guests to conveniently access resort services while helping staff and managers efficiently manage daily operations, payments, customer information, and business performance through centralized dashboards and reports.

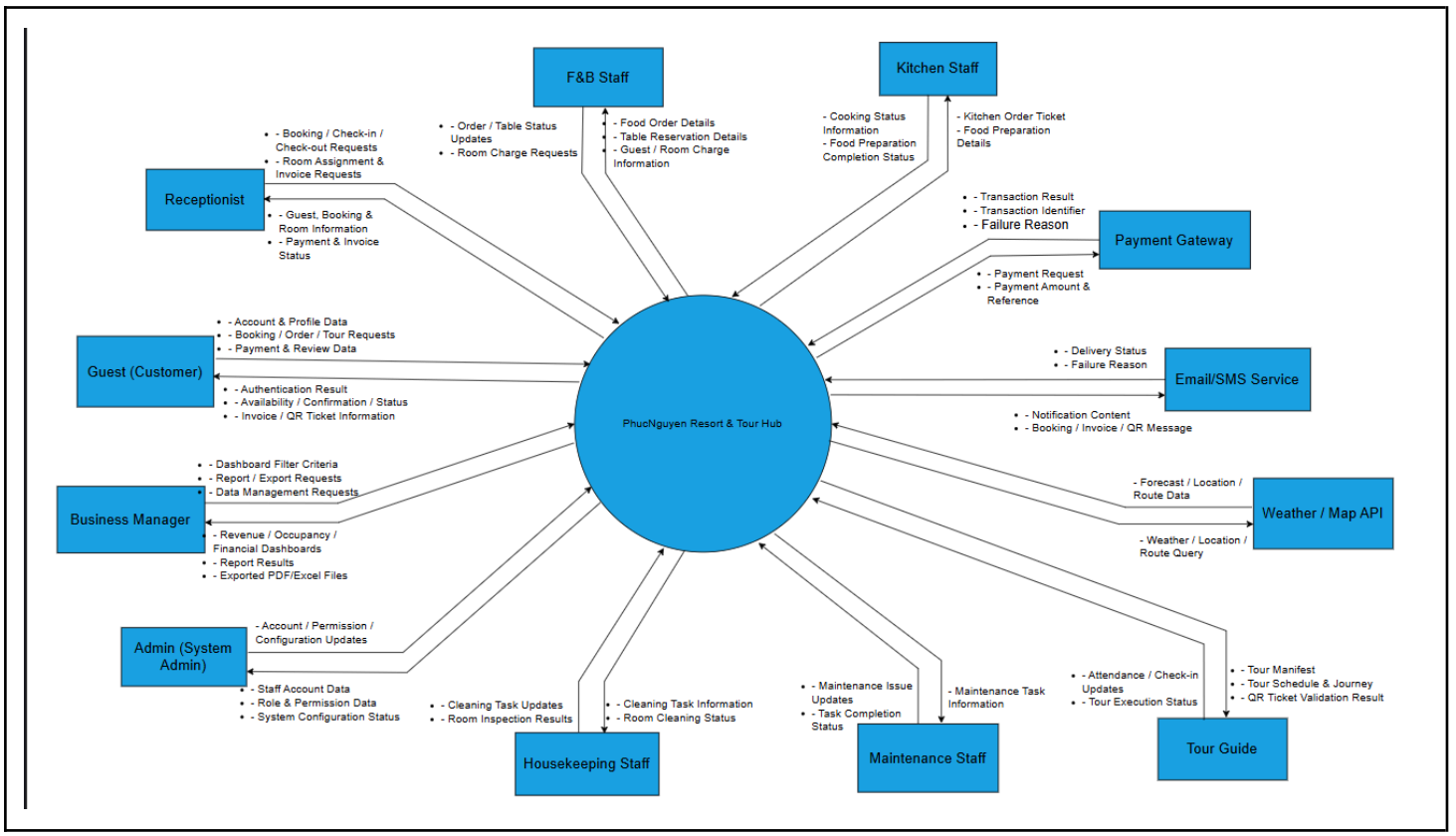


Figure I.1: Context Diagram

Context Diagram.png

[Draw.io]

From context diagram to use case modeling

2. User Requirements

2.1 Actors

#	Actor	Description
1	Guest	An unauthenticated user who accesses the system without logging in. Guests can browse resort information, search available rooms, view restaurant menus, explore tour packages, read customer reviews, and register for an account.

2	User	The user represents all external entities, specifically human beings. Users can log in, manage their personal profile, update personal information, and access authenticated system features.
3	Customer	A registered and authenticated user who actively uses resort services. Customers can book rooms, make deposit payments, reserve restaurant tables, order food, book tours, view invoices, submit reviews, and request personal data deletion.
4	Admin	The system administrator with the highest level of authority. Responsible for managing the entire system, user accounts, role assignments, account suspension/reactivation, data backup and recovery, and monitoring system activities.
5	Business Manager	The resort operations manager responsible for managing business-related data such as room categories, restaurant menus, and tour packages. The manager can monitor revenue, occupancy rates, financial reports, customer reviews, and overall business performance.
6	Receptionist	Front desk staff responsible for managing reservations, verifying guest information, performing check-in and check-out procedures, assigning rooms, updating room statuses, processing final payments, and generating consolidated invoices.
7	Housekeeping Staff	Staff responsible for maintaining room cleanliness. They can view cleaning tasks, update room cleaning statuses, mark rooms as cleaned, and report maintenance issues when necessary.
8	Maintenance Staff	Technical staff responsible for inspecting, repairing, and maintaining resort facilities and equipment. They manage maintenance requests, perform repairs, and update maintenance task statuses.
9	F&B Staff	Restaurant and room service staff responsible for handling food orders, managing table reservations, updating order statuses, delivering food, and charging food expenses to guest room accounts.
	Kitchen Staff	Kitchen personnel responsible for receiving Kitchen Order Tickets (KOTs), preparing food, updating food preparation statuses, and marking orders as ready for service.

2.2 Diagrams

2.2.1 UCs for Guest

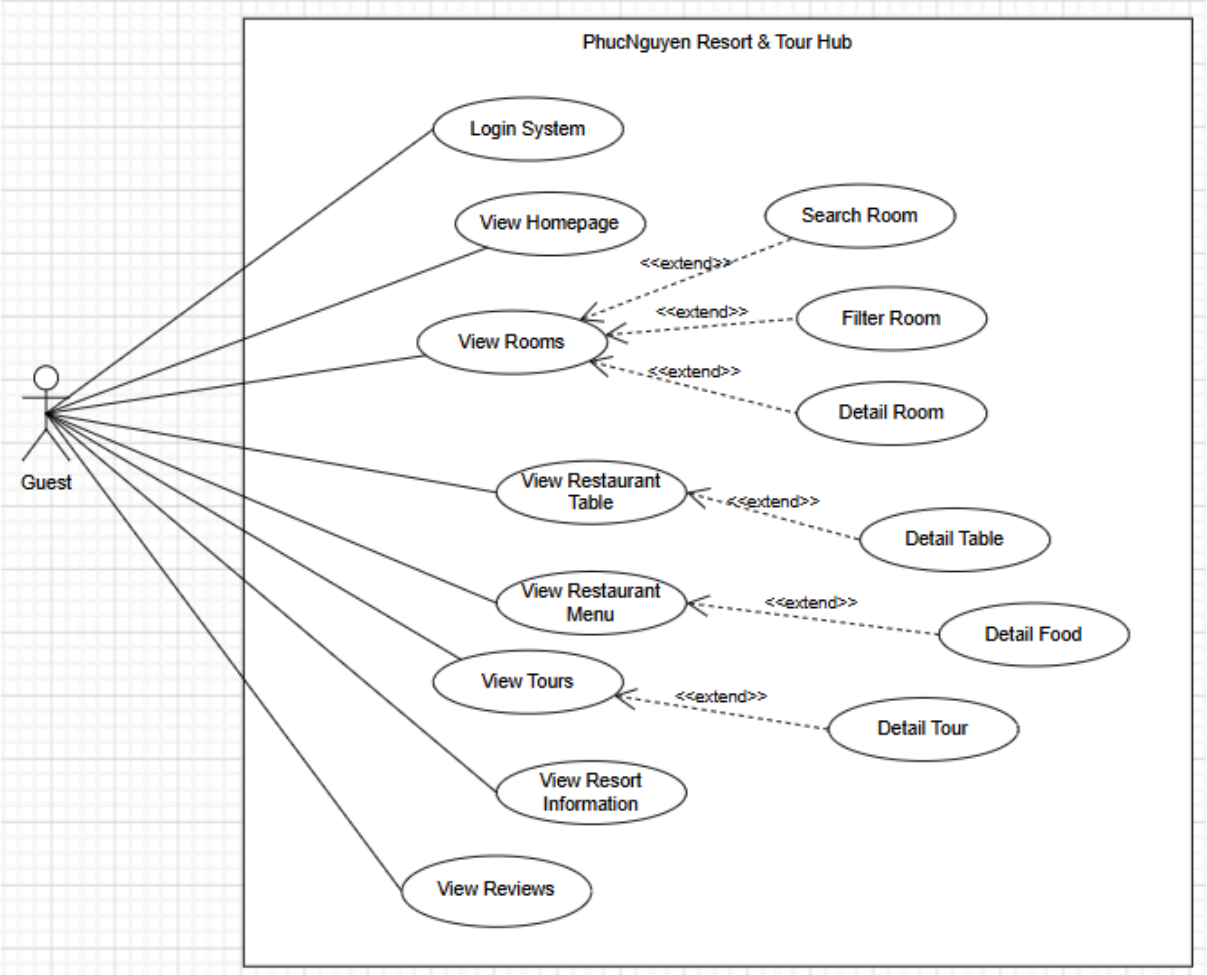


Figure I.2: Use Case Diagram - Guest

2.2.2 UCs for Registered User

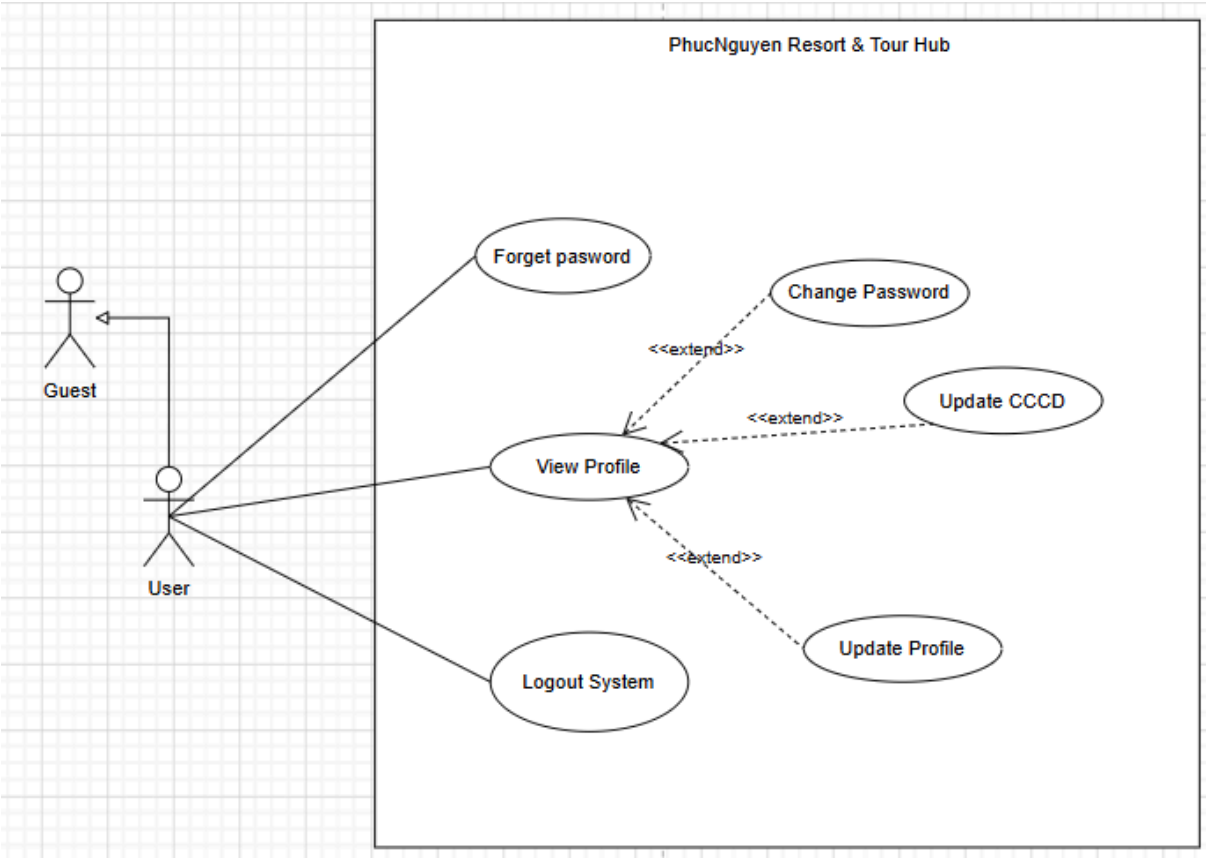


Figure I.3: Use Case Diagram -User

2.2.3 UCs for Customer

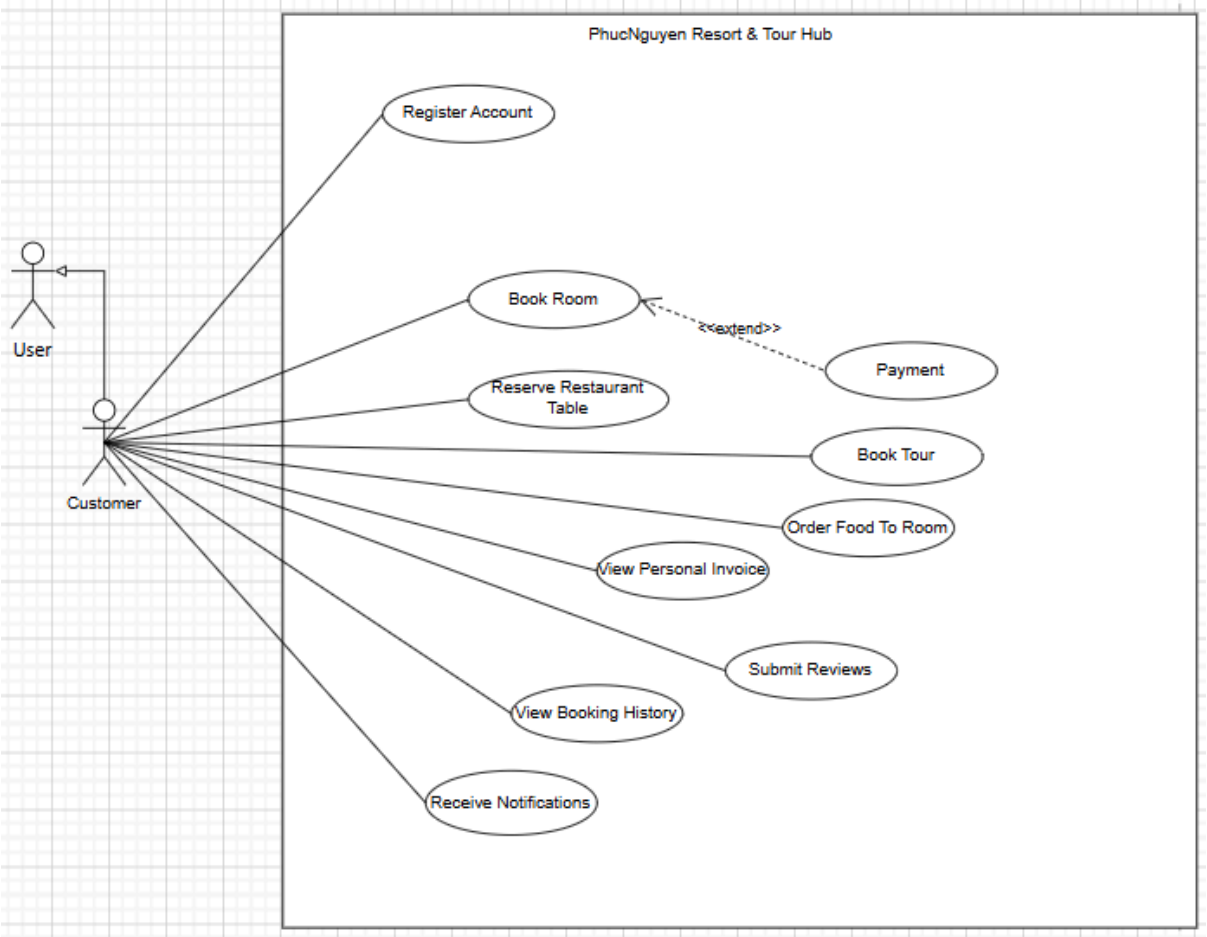


Figure I.4: Use Case Diagram - Customer

2.2.4 UCs for Business Manager:

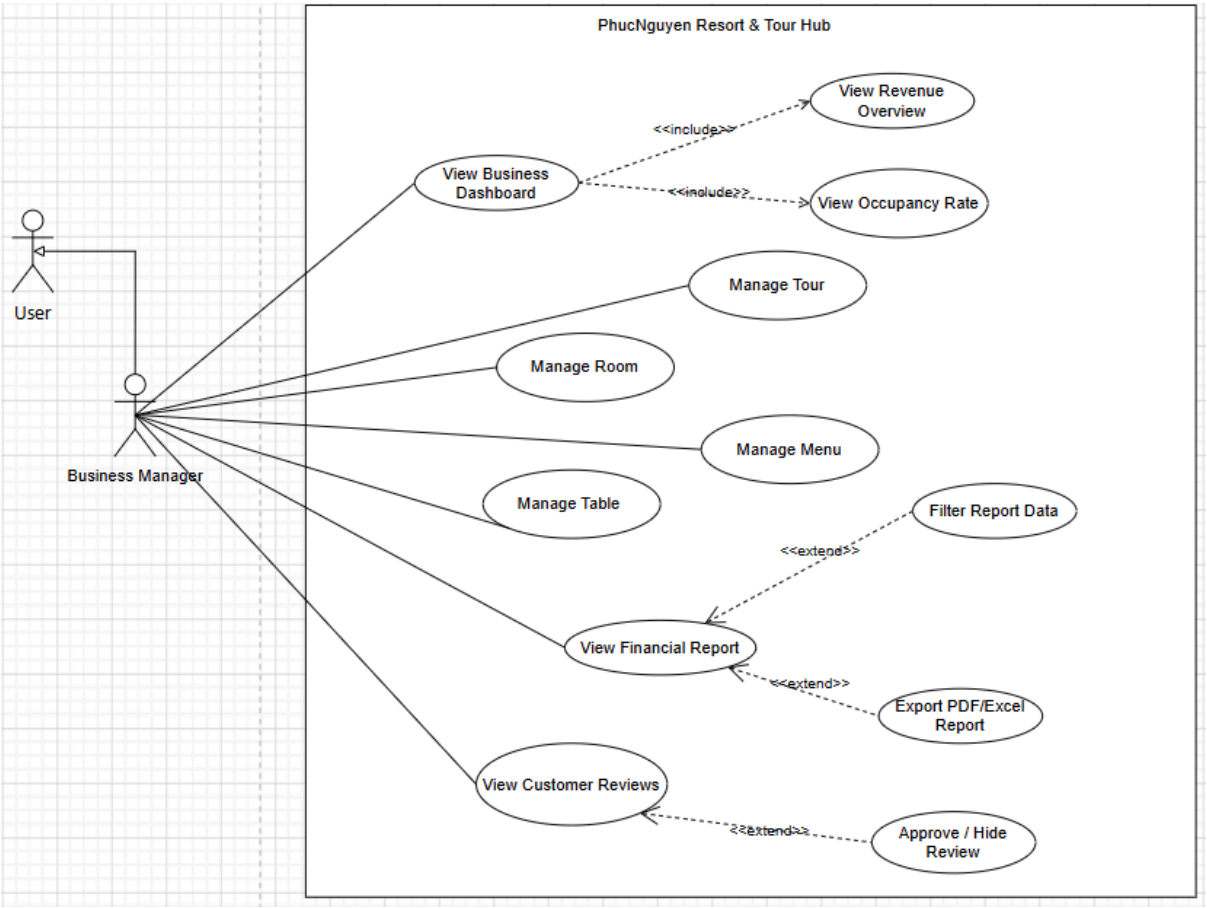


Figure I.6: Use Case Diagram -Business Manager

2.2.5 UCs for Receptionist:

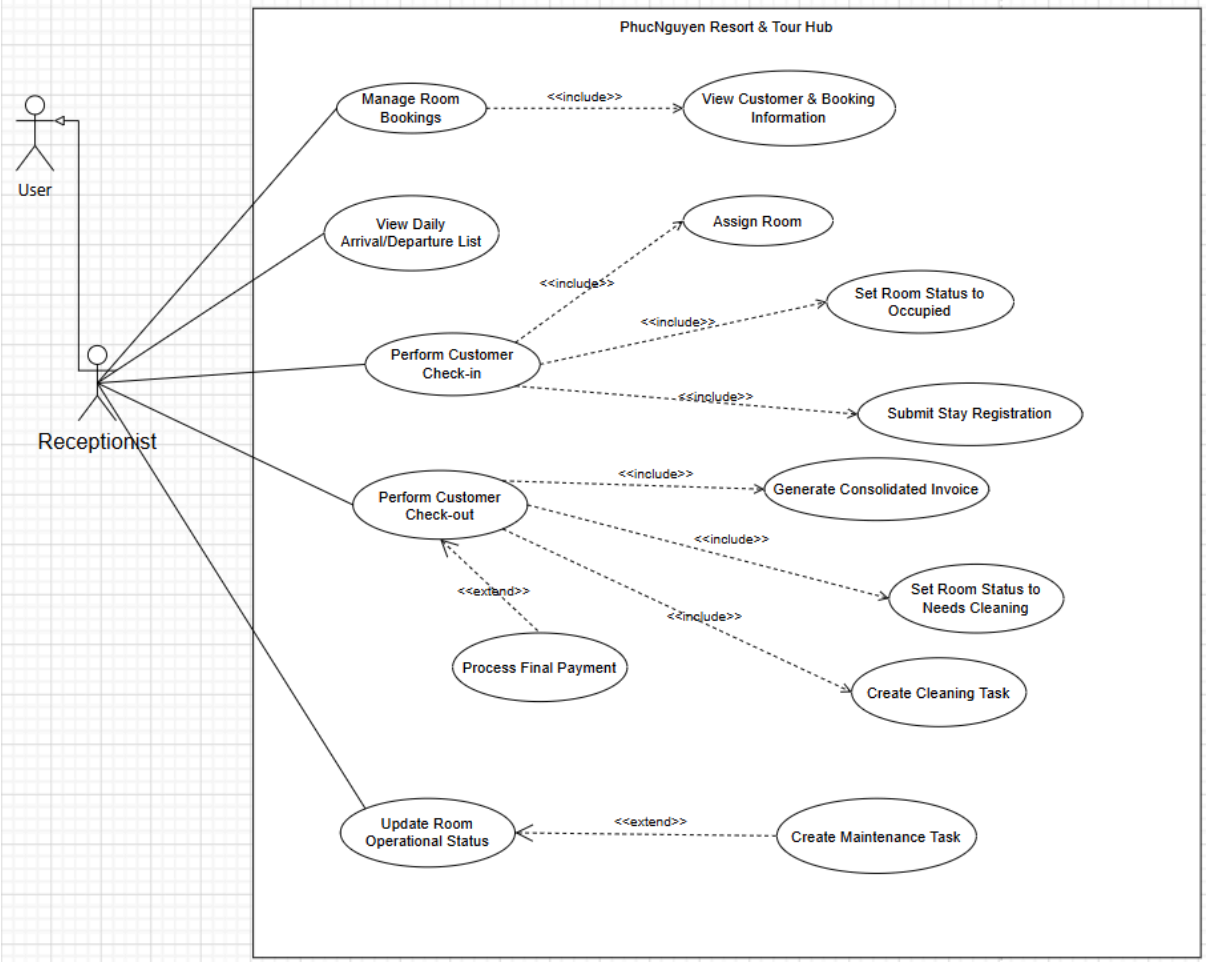


Figure I.7: Use Case Diagram - Receptionist

2.2.6 UCs for Housekeeping Staff:

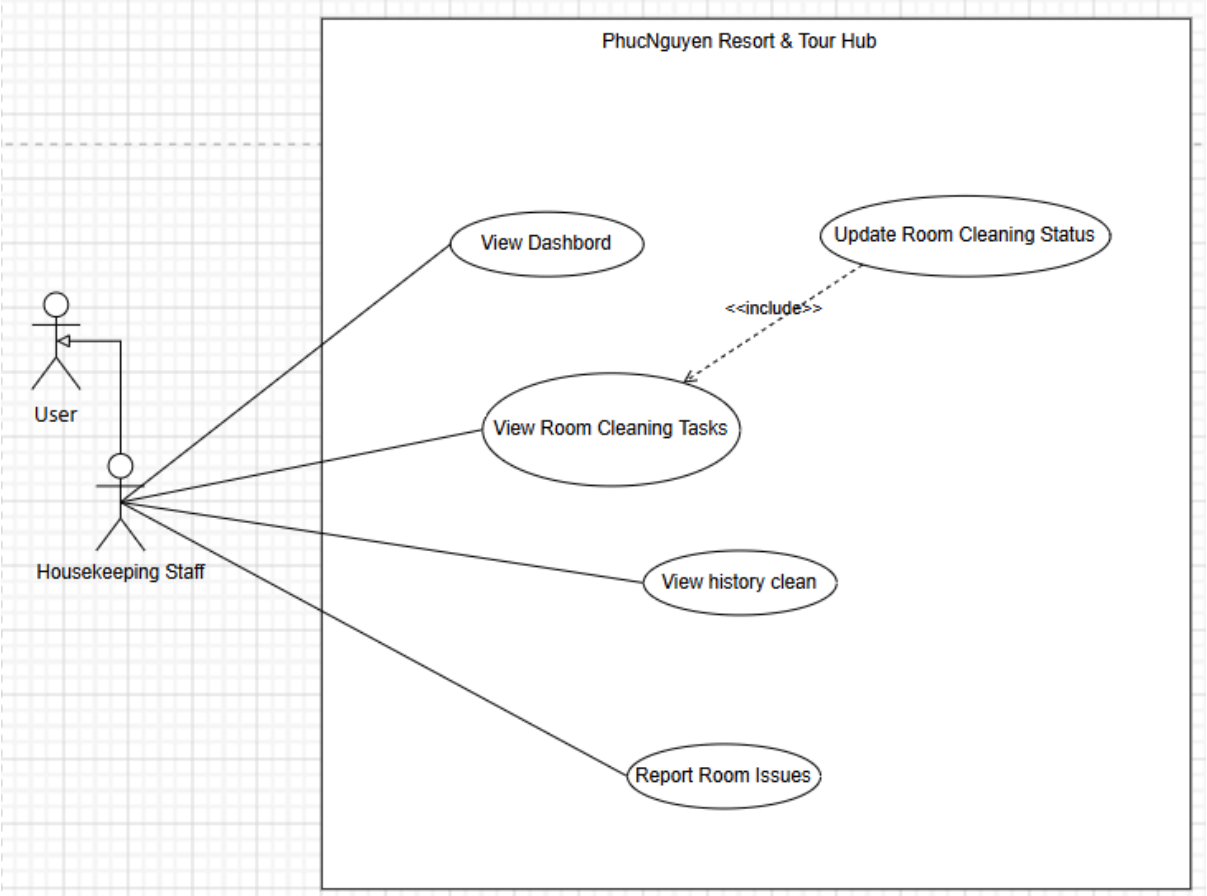


Figure I.8: Use Case Diagram - Housekeeping Staff

2.2.7 UCs for Maintenance Staff:

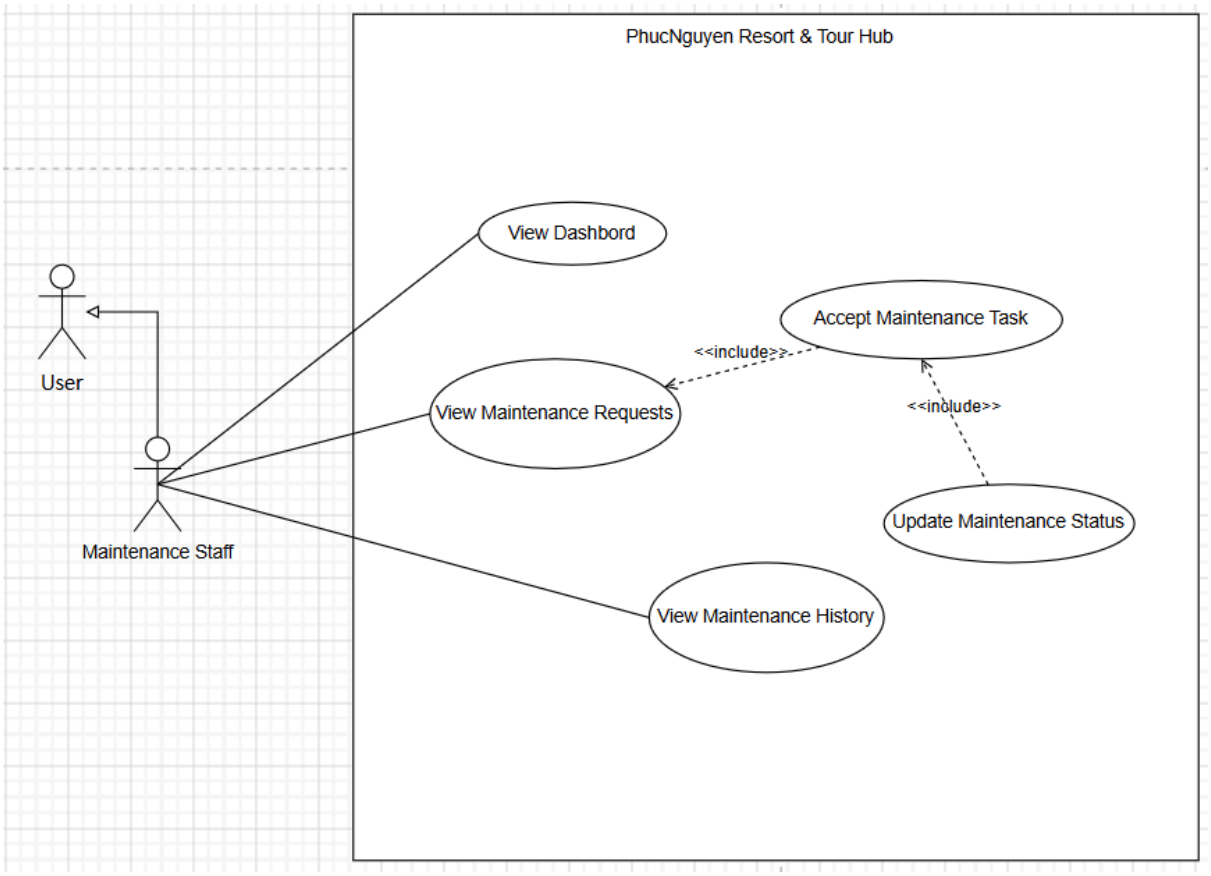


Figure I.9: Maintenance Staff

2.2.8 UCs for F&B Staff:

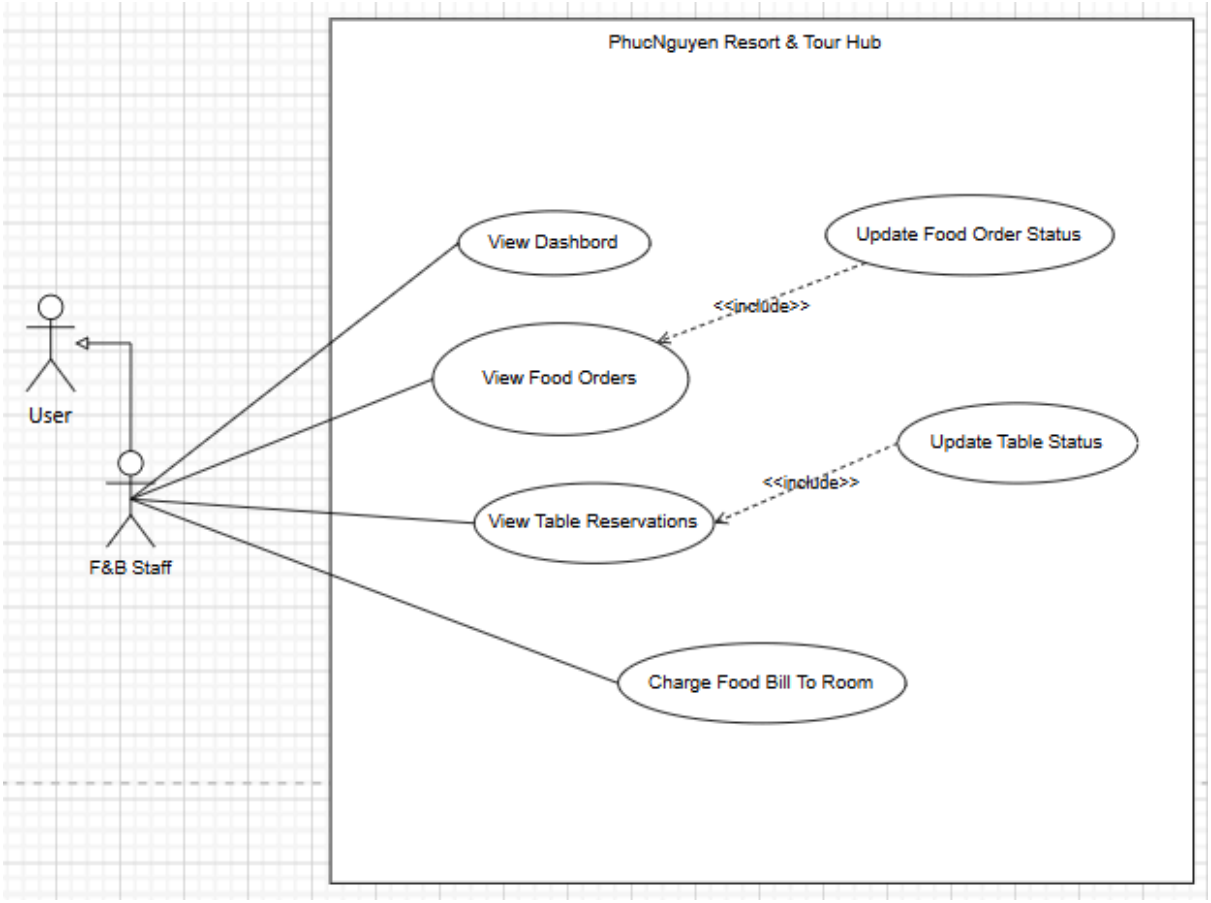


Figure I.10: F&B Staff

2.2.9 UCs for Kitchen Staff:

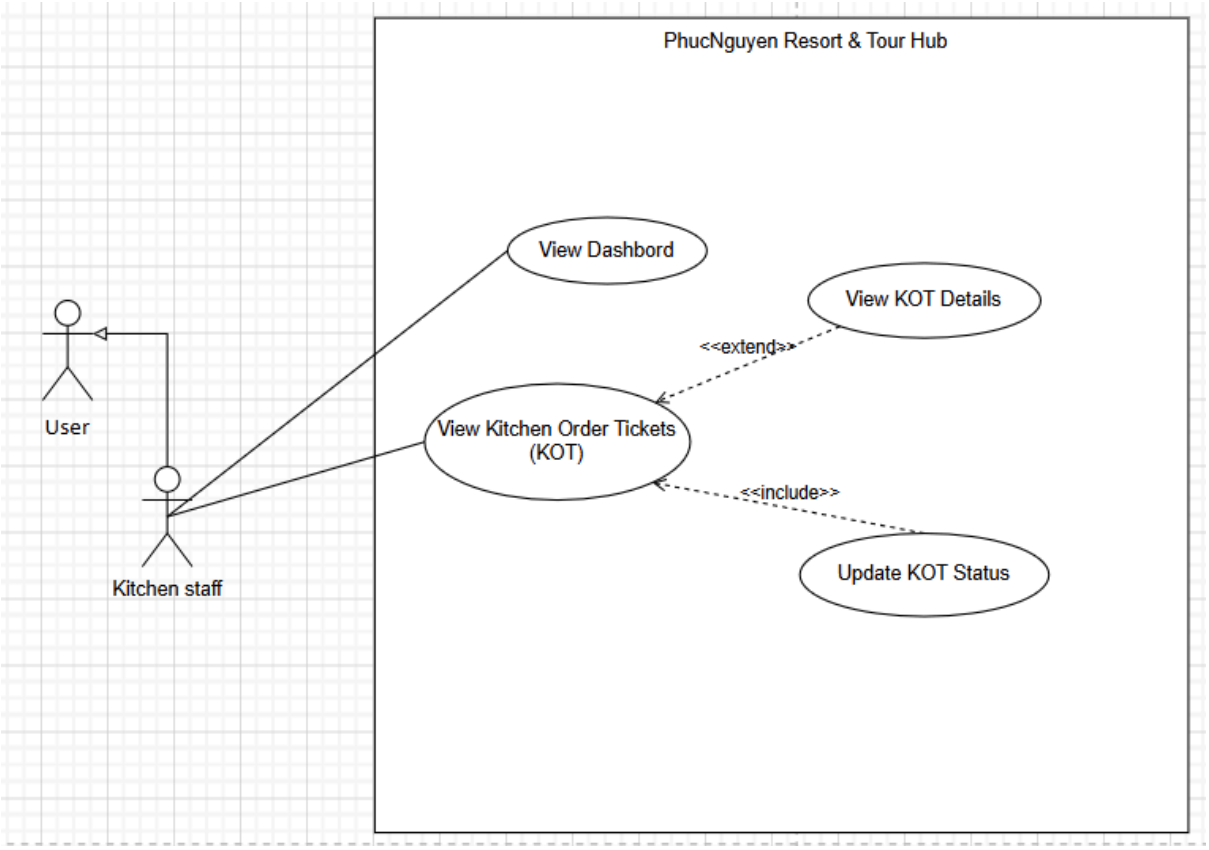


Figure I.11: Kitchen Staff

2.2.10 UCs for Admin:

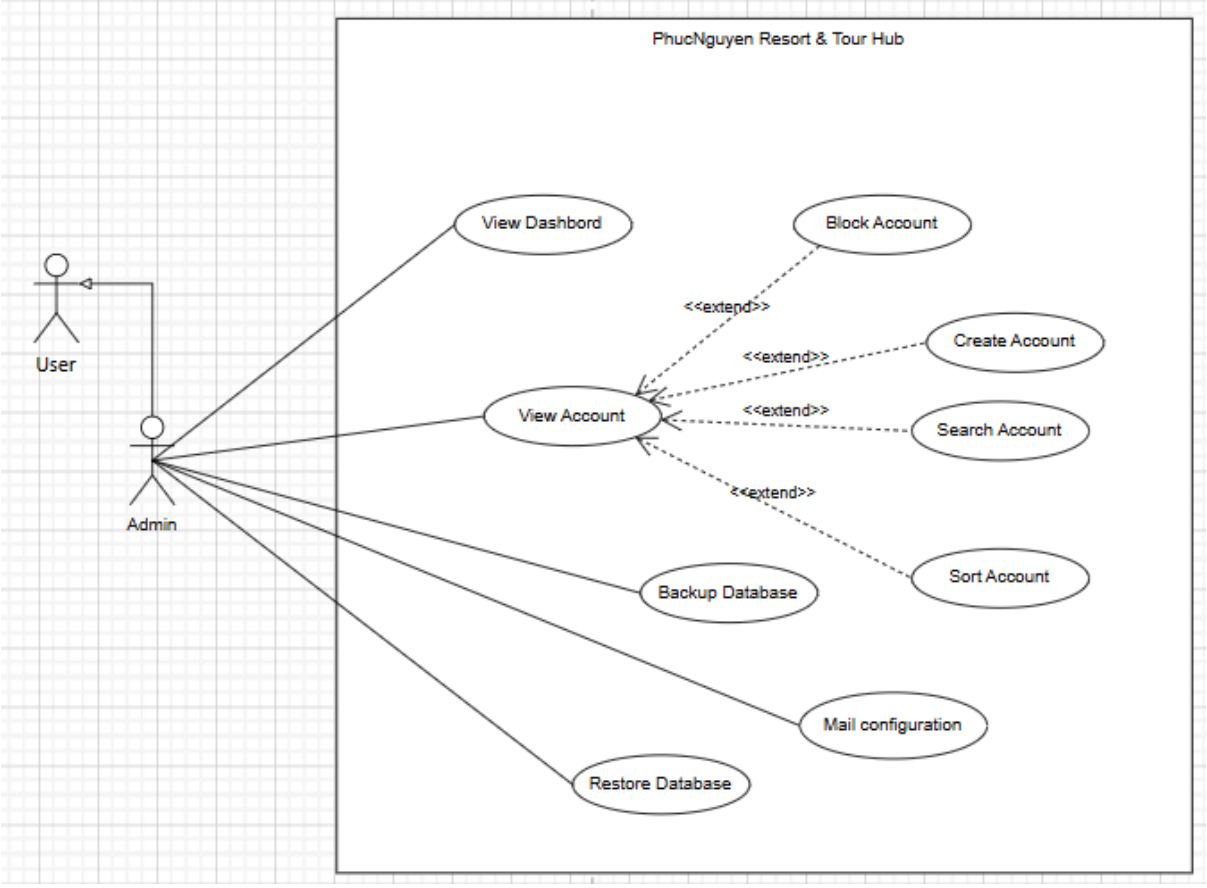


Figure I.5: Use Case Diagram - Admin

2.3 Descriptions

ID	Use Case	Actor(s)	Use Case Description
UC01	Login	Guest	As a guest, I want to log into the system so that I can access features according to my role.
UC02	View Homepage	Guest	As a guest, I want to view the homepage so that I can access resort information and available services.
UC03	View Rooms	Guest	As a guest, I want to browse available rooms so that I can find suitable accommodation.
UC04	Search Room	Guest	As a guest, I want to search rooms based on my preferences so that I can find suitable rooms quickly.
UC05	Filter Room	Guest	As a guest, I want to filter rooms by category, price, and capacity so that I can narrow down my options.
UC06	View Room Details	Guest	As a guest, I want to view room details so that I can make an informed booking decision.
UC07	View Restaurant Table	Guest	As a guest, I want to view restaurant tables so that I can reserve a table if needed.
UC08	View Table	Guest	As a guest, I want to view table details so that I can select a

	Details		suitable table.
UC09	View Restaurant Menu	Guest	As a guest, I want to browse the restaurant menu so that I can explore available dishes.
UC10	View Food Details	Guest	As a guest, I want to view food details so that I can know ingredients, prices, and descriptions.
UC11	View Tours	Guest	As a guest, I want to browse available tours so that I can plan activities during my stay.
UC12	View Tour Details	Guest	As a guest, I want to view detailed tour information so that I can decide whether to book it.
UC13	View Resort Information	Guest	As a guest, I want to view resort information so that I can learn about facilities and services.
UC14	View Reviews	Guest	As a guest, I want to view customer reviews so that I can evaluate service quality.
UC15	View Profile	Registered User	As a registered user, I want to view my profile information so that I can verify my account details.
UC16	Update Profile	Registered User	As a registered user, I want to update my profile information so that my account information remains accurate.
UC17	Update CCCD/Passport	Registered User	As a registered user, I want to upload or update my identification documents for verification purposes.
UC18	Change Password	Registered User	As a registered user, I want to change my password so that I can secure my account.
UC19	Forgot Password	Registered User	As a registered user, I want to reset my password when I forget it so that I can regain access to my account.
UC20	Logout	Registered User	As a registered user, I want to log out of the system so that my account remains secure.
UC21	Register Account	Customer	As a customer, I want to create an account so that I can use resort services.
UC22	Book Room	Customer	As a customer, I want to book a room so that I can stay at the resort.
UC23	Payment	Customer	As a customer, I want to pay deposits and service fees online so that my bookings can be confirmed.
UC24	Reserve Restaurant Table	Customer	As a customer, I want to reserve a restaurant table so that I can dine at my preferred time.
UC25	Order Food To Room	Customer	As a customer, I want to order food to my room so that I can enjoy room service.
UC26	Book Tour	Customer	As a customer, I want to book a tour so that I can participate in resort activities.
UC27	View Personal Invoice	Customer	As a customer, I want to view my invoices so that I can track my expenses.

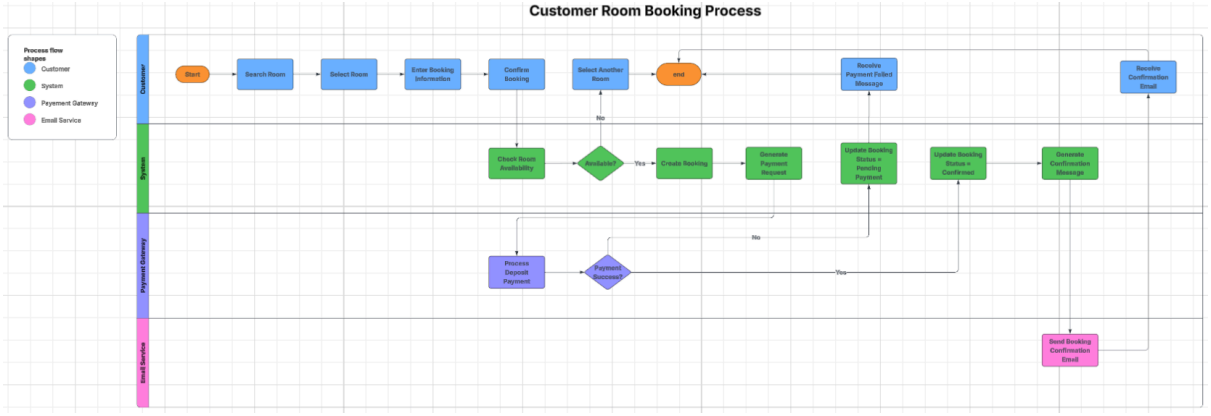
UC28	View Booking History	Customer	As a customer, I want to view my booking history so that I can review past reservations.
UC29	Submit Reviews	Customer	As a customer, I want to submit reviews so that I can share my experience.
UC30	Receive Notifications	Customer	As a customer, I want to receive notifications so that I can stay updated about bookings and services.
UC31	View Dashboard	Admin	As an administrator, I want to view the system dashboard so that I can monitor system activities.
UC32	View Account	Admin	As an administrator, I want to manage user accounts so that I can control system access.
UC33	Create Account	Admin	As an administrator, I want to create staff accounts so that employees can access the system.
UC34	Search Account	Admin	As an administrator, I want to search user accounts so that I can manage them efficiently.
UC35	Sort Account	Admin	As an administrator, I want to sort accounts so that I can find information quickly.
UC36	Block Account	Admin	As an administrator, I want to block accounts that violate system policies.
UC37	Backup Database	Admin	As an administrator, I want to back up the database so that system data is protected.
UC38	Restore Database	Admin	As an administrator, I want to restore the database so that data can be recovered when needed.
UC39	View Business Dashboard	Business Manager	As a business manager, I want to view business dashboards so that I can monitor resort performance.
UC40	View Revenue Overview	Business Manager	As a business manager, I want to view revenue summaries so that I can evaluate business income.
UC41	View Occupancy Rate	Business Manager	As a business manager, I want to view occupancy rates so that I can monitor room utilization.
UC42	Monitor Resort Performance	Business Manager	As a business manager, I want to monitor operational performance so that I can improve efficiency.
UC43	Analyze Revenue by Service	Business Manager	As a business manager, I want to analyze revenue sources so that I can identify profitable services.
UC44	View Financial Report	Business Manager	As a business manager, I want to view financial reports so that I can evaluate business results.
UC45	Filter Report Data	Business Manager	As a business manager, I want to filter report data so that I can focus on specific information.
UC46	Export PDF/Excel Report	Business Manager	As a business manager, I want to export reports so that I can share and archive business information.
UC47	Moderate	Business Manager	As a business manager, I want to manage customer reviews

	Customer Reviews		so that inappropriate content can be controlled.
UC48	View Customer Reviews	Business Manager	As a business manager, I want to review customer feedback so that I can improve service quality.
UC49	Approve/Hide Review	Business Manager	As a business manager, I want to approve or hide reviews so that displayed feedback remains appropriate.
UC50	Manage Room Bookings	Receptionist	As a receptionist, I want to manage room bookings so that reservations are handled properly.
UC51	View Customer & Booking Information	Receptionist	As a receptionist, I want to view guest information so that I can verify reservations.
UC52	View Daily Arrival/Departure List	Receptionist	As a receptionist, I want to view daily arrivals and departures so that I can prepare operations.
UC53	Perform Customer Check-in	Receptionist	As a receptionist, I want to check guests in so that they can begin their stay.
UC54	Assign Room	Receptionist	As a receptionist, I want to assign rooms to guests during check-in.
UC55	Submit Stay Registration	Receptionist	As a receptionist, I want to submit guest stay registration information to comply with regulations.
UC56	Perform Customer Check-out	Receptionist	As a receptionist, I want to check guests out so that their stay can be completed.
UC57	Generate Consolidated Invoice	Receptionist	As a receptionist, I want to generate consolidated invoices including room, food, and tour charges.
UC58	Process Final Payment	Receptionist	As a receptionist, I want to process final payments so that invoices can be settled.
UC59	Update Room Operational Status	Receptionist	As a receptionist, I want to update room statuses so that room availability is accurate.
UC60	Create Cleaning Task	Receptionist	As a receptionist, I want to create cleaning tasks after guest check-out.
UC61	Create Maintenance Task	Receptionist	As a receptionist, I want to create maintenance requests when room issues are reported.
UC62	View Dashboard	Housekeeping Staff	As housekeeping staff, I want to view my dashboard so that I can monitor assigned tasks.
UC63	View Room Cleaning Tasks	Housekeeping Staff	As housekeeping staff, I want to view cleaning tasks so that I know which rooms require cleaning.

UC64	Update Room Cleaning Status	Housekeeping Staff	As housekeeping staff, I want to update cleaning status so that room availability is accurate.
UC65	View Cleaning History	Housekeeping Staff	As housekeeping staff, I want to view cleaning history so that I can track completed tasks.
UC66	Report Room Issues	Housekeeping Staff	As housekeeping staff, I want to report room issues so that maintenance staff can resolve them.
UC67	View Dashboard	F&B Staff	As an F&B Staff, I want to view the operational dashboard so that I can monitor food orders, table reservations, and restaurant activities in real time.
UC68	View Food Orders	F&B Staff	As an F&B Staff, I want to view all food orders so that I can track customer requests and manage the order workflow efficiently.
UC69	Update Food Order Status	F&B Staff	As an F&B Staff, I want to update the status of food orders (Pending → Cooking → Delivered) so that customers and other staff can track order progress.
UC70	View Table Reservations	F&B Staff	As an F&B Staff, I want to view table reservations so that I can prepare tables and manage restaurant seating arrangements.
UC71	Update Table Status	F&B Staff	As an F&B Staff, I want to update table status (Available, Reserved, Occupied) so that reservation information remains accurate.
UC72	Charge Food Bill To Room	F&B Staff	As an F&B Staff, I want to charge a customer's food bill directly to their room so that the amount can be included in the final consolidated invoice.
UC73	View Dashboard	Kitchen Staff	As a Kitchen Staff member, I want to view the kitchen dashboard so that I can monitor incoming orders and kitchen workload.
UC74	View Kitchen Order Tickets (KOT)	Kitchen Staff	As a Kitchen Staff member, I want to view all Kitchen Order Tickets (KOT) so that I can prepare food according to customer requests.
UC75	View KOT Details	Kitchen Staff	As a Kitchen Staff member, I want to view detailed information of a KOT so that I can prepare the correct dishes and quantities.
UC76	Update KOT Status	Kitchen Staff	As a Kitchen Staff member, I want to update the status of a KOT (Pending → Preparing → Ready to Serve) so that F&B Staff can know when orders are ready for delivery.

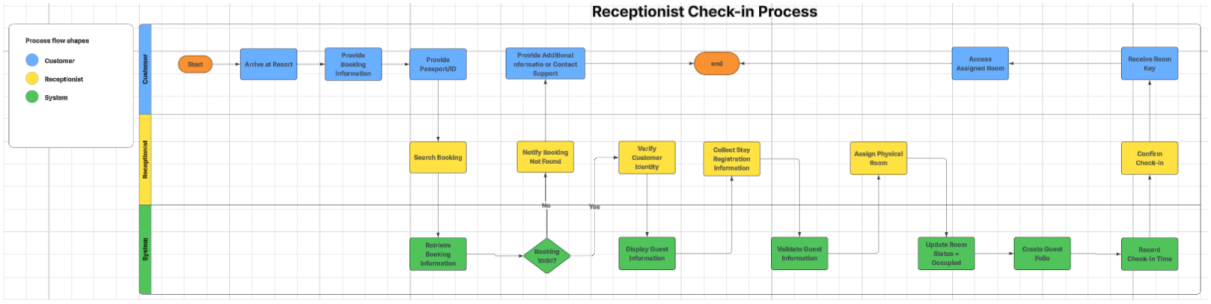
3. Activity Diagram

3.1 Customer Room Booking Process



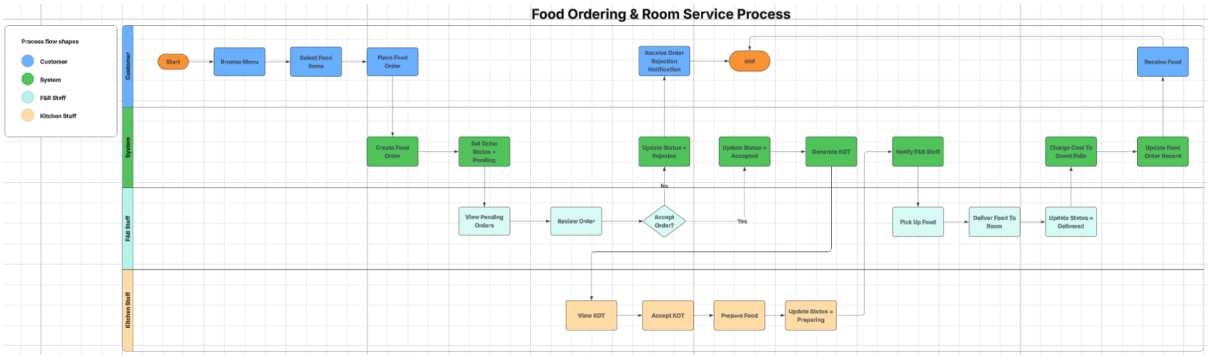
booking.png

3.2 Receptionist Check-in Process



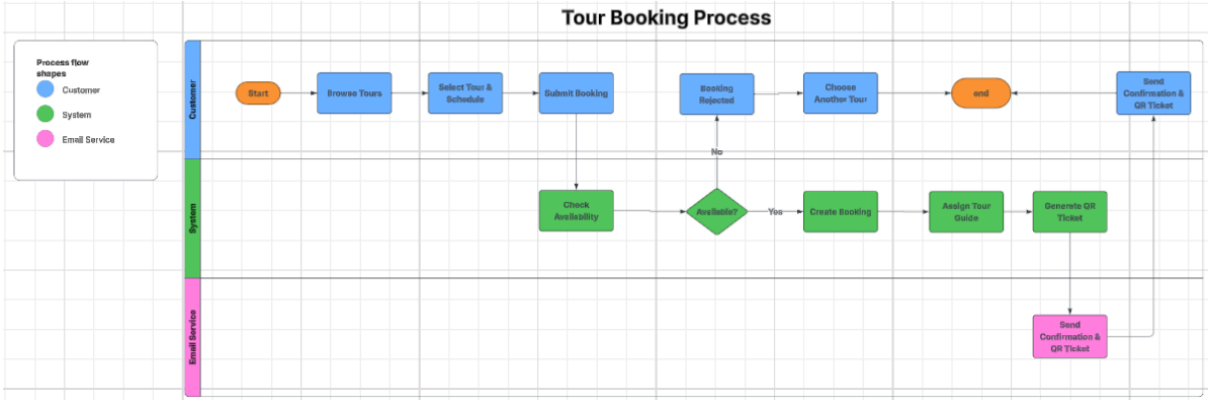
checkin.png

3.3 Food Ordering & Room Service Process



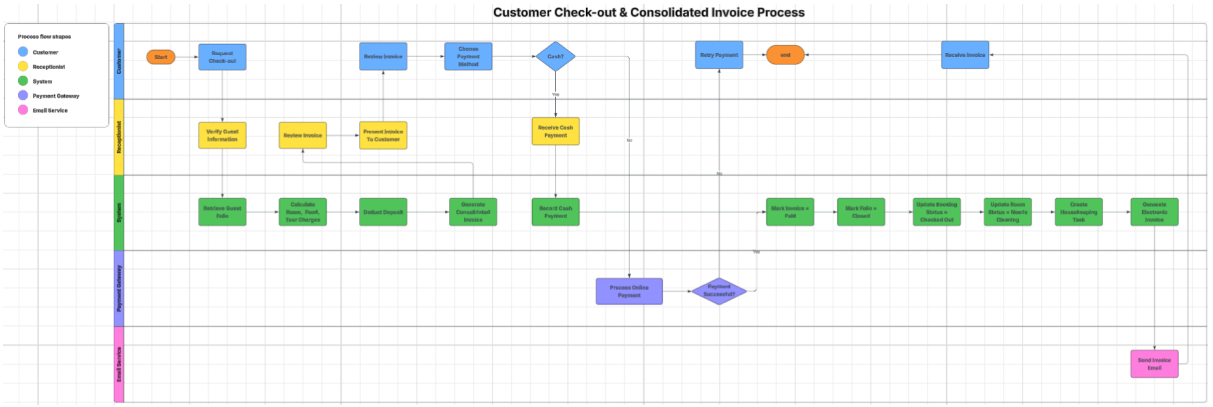
foodorder.png

3.4 Tour Booking Process



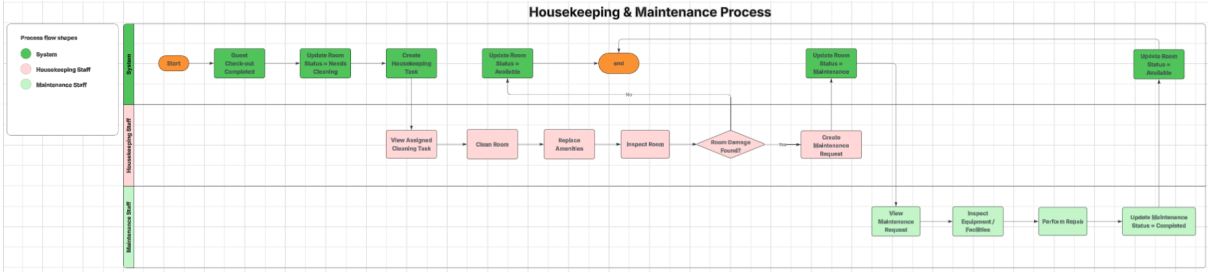
tourbooking.png

3.5 Customer Check-out & Consolidated Invoice Process



checkout.png

3.6 Housekeeping & Mai ntenance Process

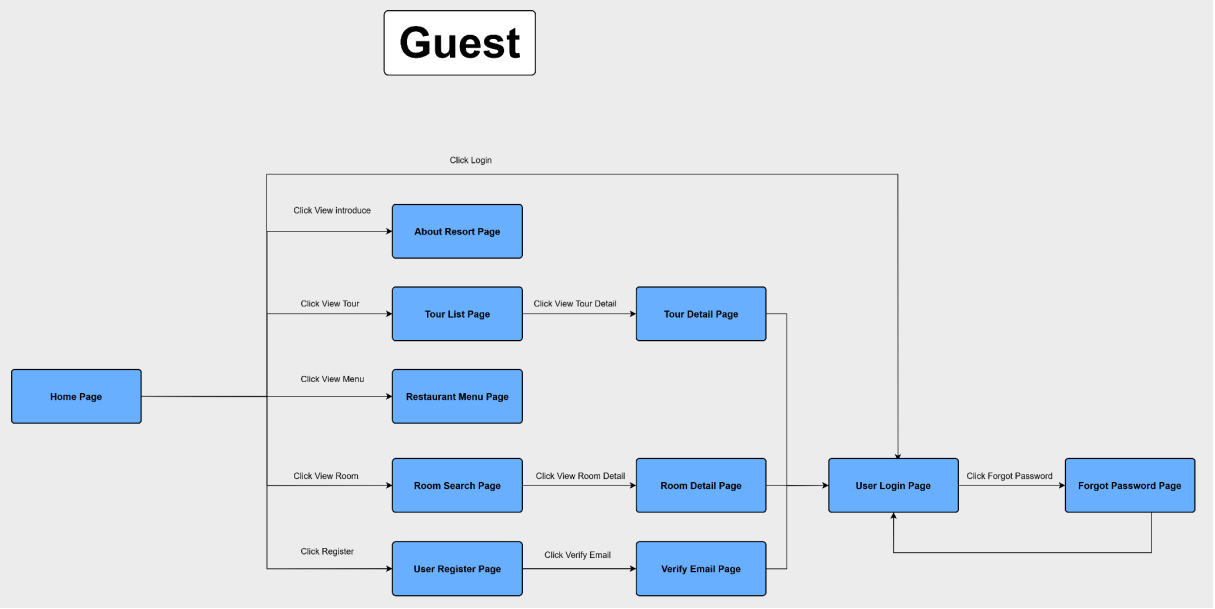


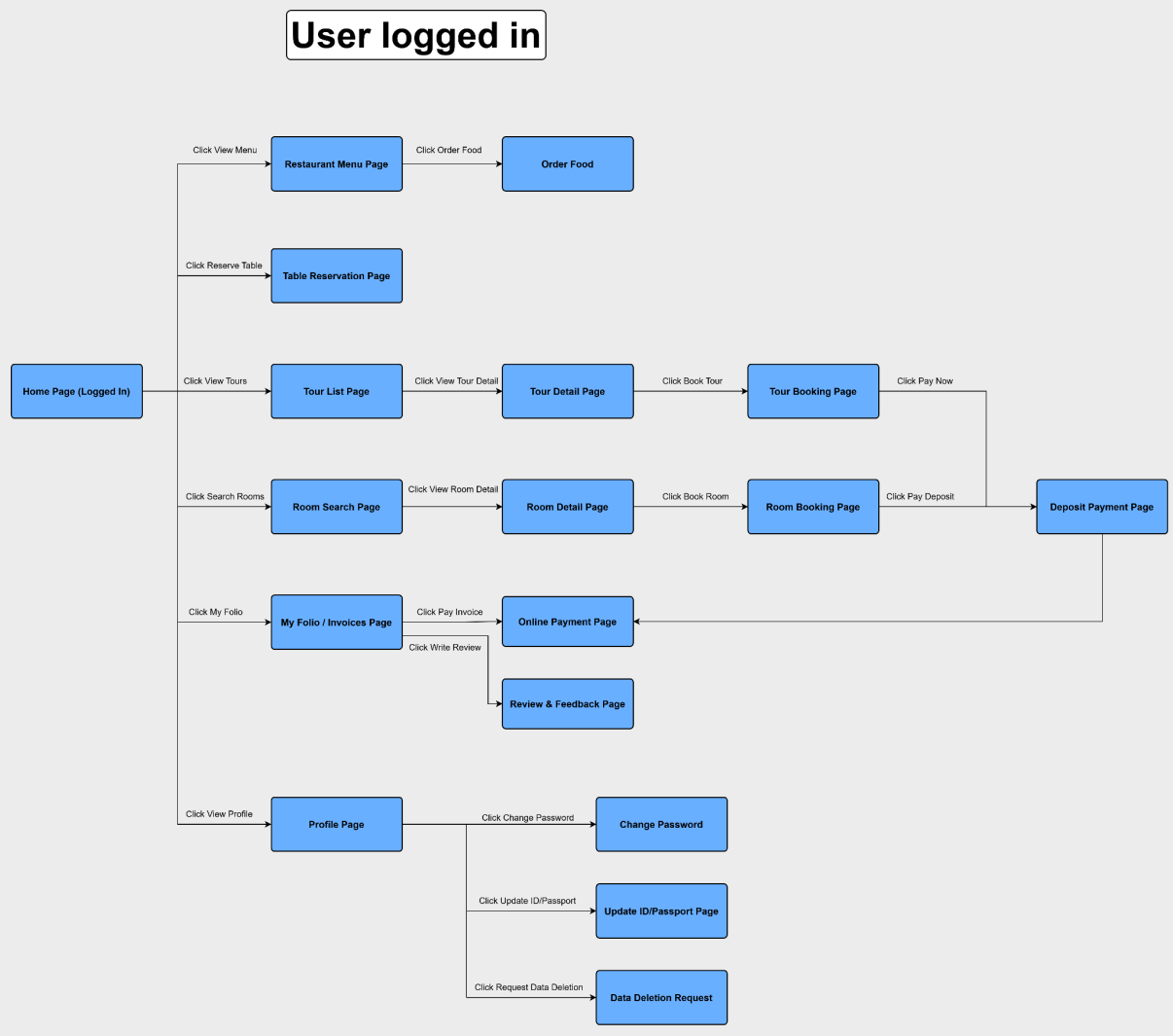
cleanroom.png

4. System Functionalities

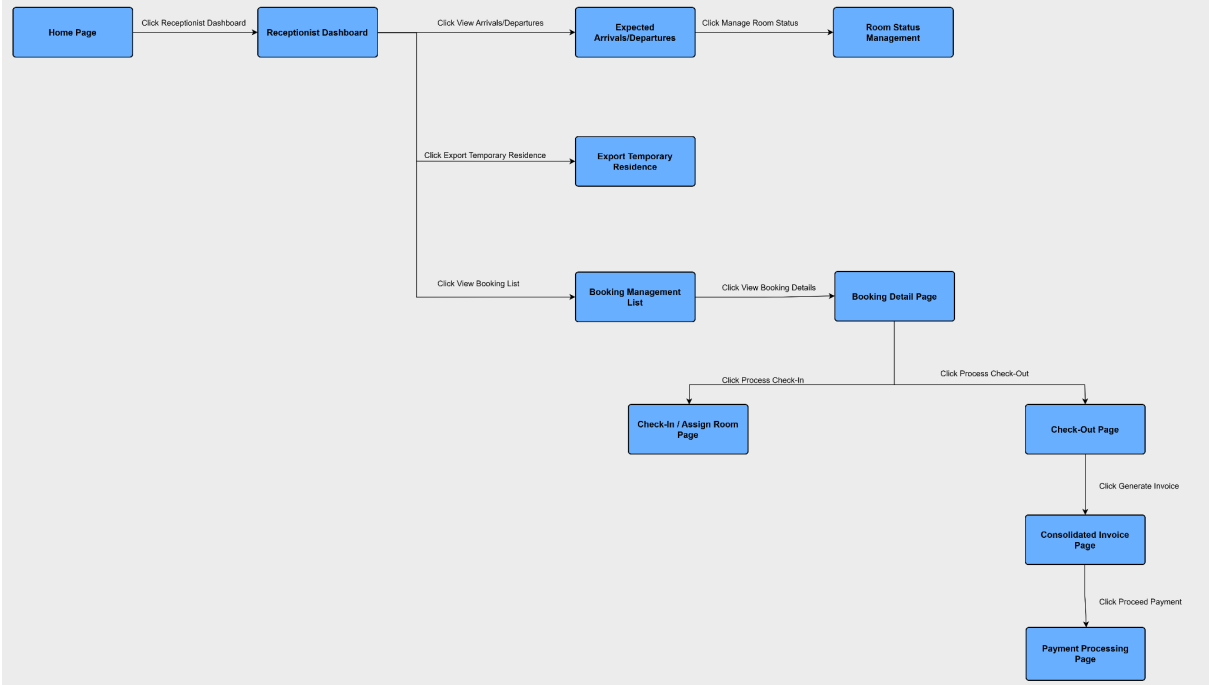
4.1 Screens Flow

4.1.1 Guest Screen Flow:

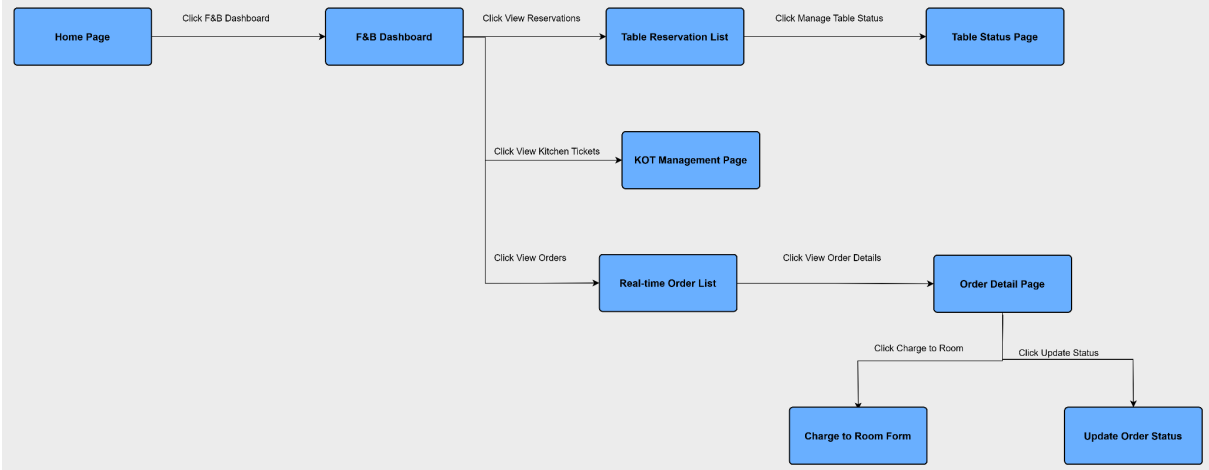



User logged in
Home Page (Logged In)
Click View Menu
Restaurant Menu Page
Click Order Food
Order Food
Click Reserve Table
Table Reservation Page
Click View Tours
Tour List Page
Click View Tour Detail
Tour Detail Page
Click Book Tour
Tour Booking Page
Click Pay Now
Click Search Rooms
Room Search Page
Click View Room Detail
Room Detail Page
Click Book Room
Room Booking Page
Click Pay Deposit
Deposit Payment Page
Click My Folio
My Folio / Invoices Page
Click Pay Invoice
Online Payment Page
Click Write Review
Review & Feedback Page
Click View Profile
Profile Page
Click Change Password
Change Password
Click Update ID/Passport
Update ID/Passport Page
Click Request Data Deletion
Data Deletion Request

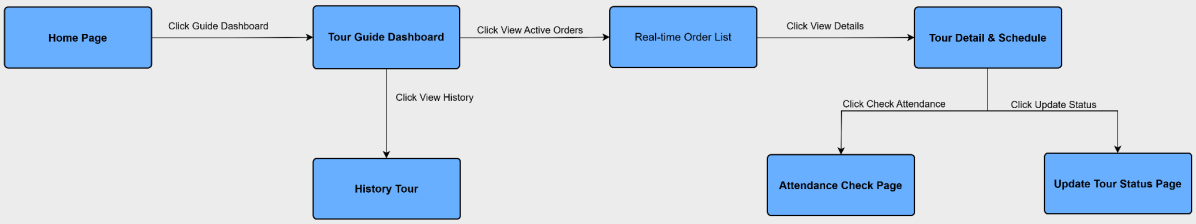
Receptionist



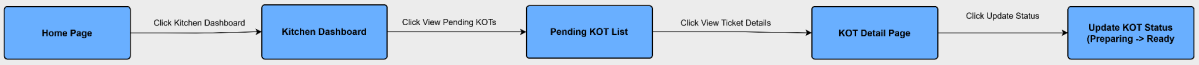
F&B Staff



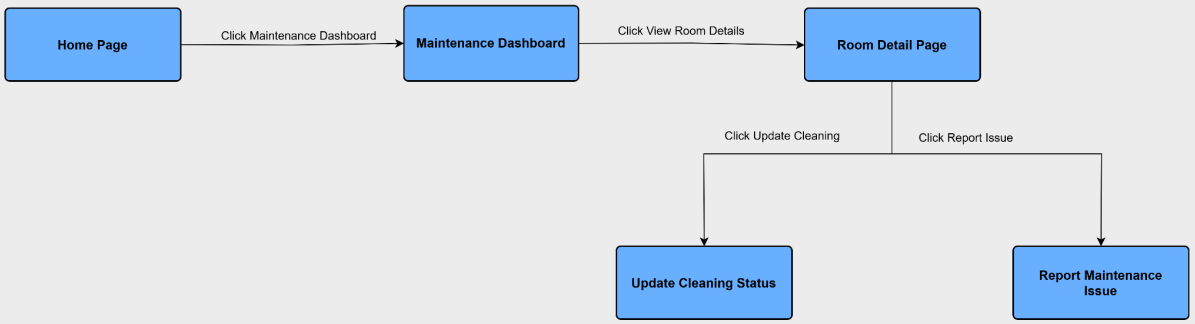
Tour Guide



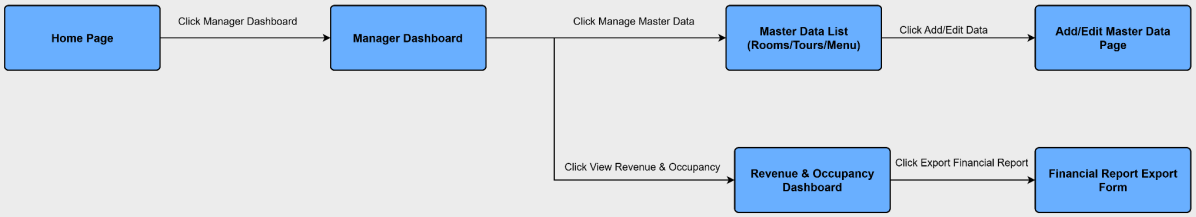
Kitchen Staff

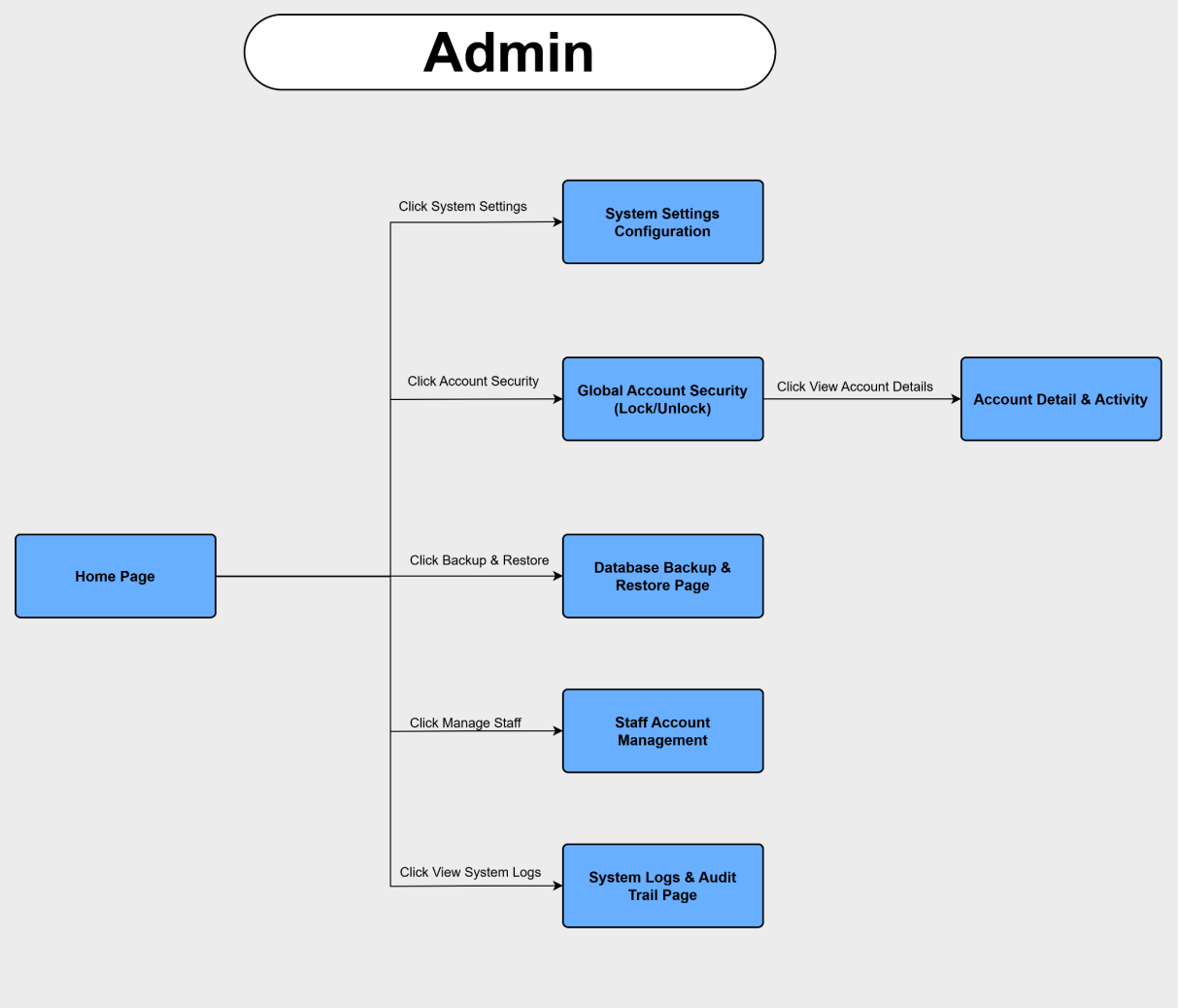


Maintenance



Manager





4.2 Screen Authorization

Screen Name	Guest	User	Tour Guide	Manager	Receptionist	F&B Staff	Kitchen Staff	Maintenance	Admin
Home Page	✓	✓	✓	✓	✓	✓	✓	✓	✓
User Login Page	✓	✓	✓	✓	✓	✓	✓	✓	✓
User Register Page	✓	✓							
Verify Email Page	✓	✓							
Forgot Password Page	✓	✓							
About Resort Page	✓	✓	✓	✓	✓	✓	✓	✓	✓

Tour List Page	✓	✓							
Tour Detail Page	✓	✓							
Restaurant Menu Page	✓	✓							
Room Search Page	✓	✓							
Room Detail Page	✓	✓						✓	
Profile Page		✓	✓	✓	✓	✓	✓	✓	✓
Change Password		✓							
Update ID/Passport Page		✓							
Data Deletion Request		✓							
Order Food		✓							
Table Reservation Page		✓							
Tour Booking Page		✓							
Room Booking Page		✓							
Deposit Payment Page		✓							
My Folio / Invoices Page		✓							
Online Payment Page		✓							
Review & Feedback Page		✓							
Tour Guide Dashboard			✓						

History Tour			✓						
Real-time Order List			✓			✓			
Tour Detail & Schedule			✓						
Attendance Check Page			✓						
Update Tour Status Page			✓						
Manager Dashboard				✓					
Master Data List (Rooms/Tours/ Menu)				✓					
Add/Edit Master Data Page				✓					
Revenue & Occupancy Dashboard				✓					
Financial Report Export Form				✓					
Staff Account Management									✓
Receptionist Dashboard					✓				
Expected Arrivals/Depart ures					✓				
Export Temporary Residence					✓				
Booking Management List					✓				

Booking Detail Page					✓				
Room Status Management					✓				
Check-In / Assign Room Page					✓				
Check-Out Page					✓				
Consolidated Invoice Page					✓				
Payment Processing Page					✓				
F&B Dashboard						✓			
Table Reservation List						✓			
Table Status Page						✓			
KOT Management Page						✓			
Order Detail Page						✓			
Charge to Room Form						✓			
Update Order Status						✓			
Kitchen Dashboard							✓		
Pending KOT List							✓		
KOT Detail Page							✓		
Update KOT Status							✓		

(Preparing -> Ready)									
Maintenance Dashboard								✓	
Update Cleaning Status								✓	
Report Maintenance Issue								✓	
System Settings Configuration									✓
Global Account Security (Lock/Unlock)									✓
Account Detail & Activity									✓
Database Backup & Restore Page									✓
System Logs & Audit Trail Page									✓

4.3 Non-UI Functions

#	Feature	System Function	Description

II. Functional Requirements

III. System Design

1. Database Design

1.1 Database Schema

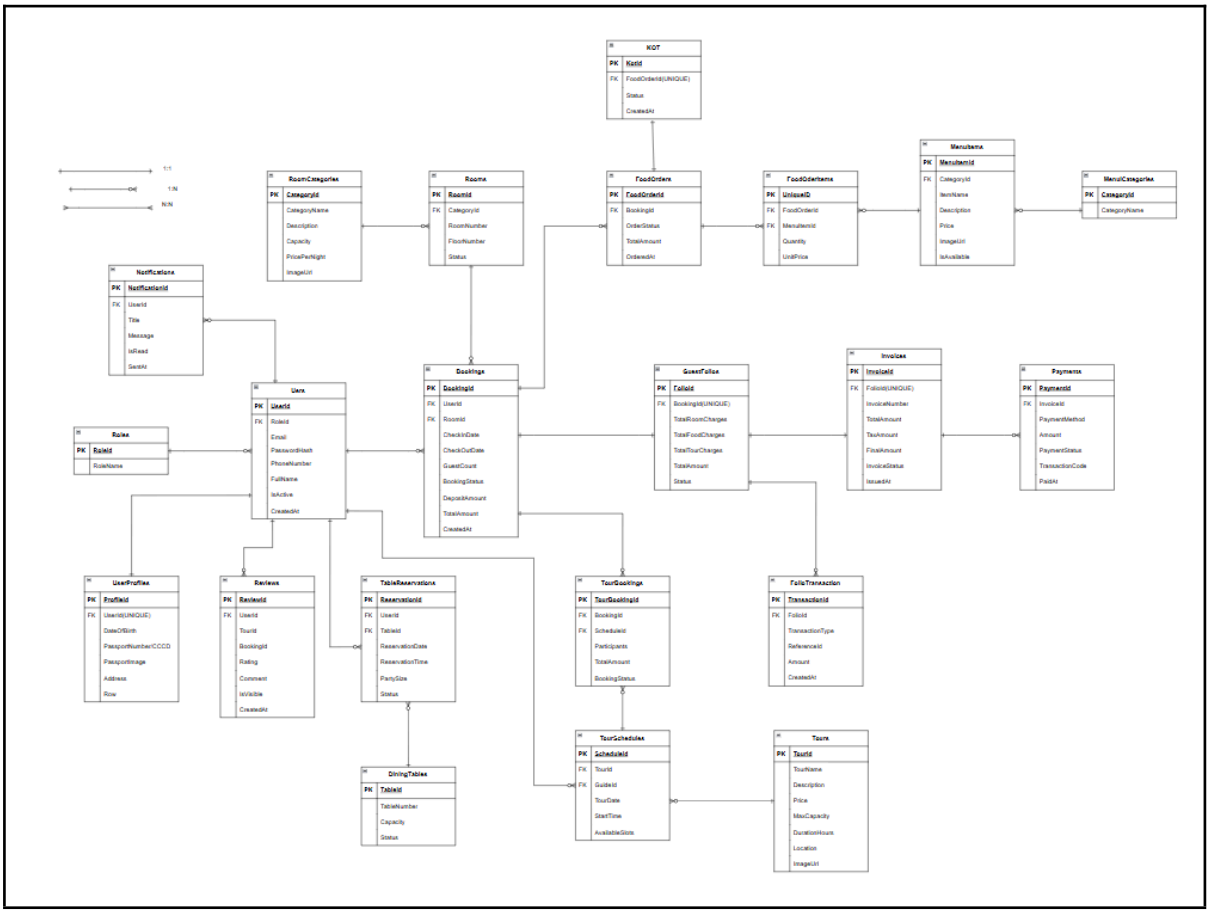


Figure III.1: Entity Relationship Diagram

draw

1.2 Table Descriptions

No	Table	Description
1	Roles	• Primary keys: RoleId • Foreign keys: None • Unique keys: RoleName
2	Users	• Primary keys: UserId • Foreign keys: RoleId → Roles(RoleId) • Unique keys: Email
3	UserProfiles	• Primary keys: ProfileId • Foreign keys: UserId → Users(UserId) • Unique keys: UserId
4	RoomCategories	• Primary keys: CategoryId • Foreign keys: None • Unique keys: None

5	Rooms	• Primary keys: RoomId • Foreign keys: CategoryId → RoomCategories(CategoryId) • Unique keys: RoomNumber
6	Bookings	• Primary keys: BookingId • Foreign keys: UserId → Users(UserId), RoomId → Rooms(RoomId) • Unique keys: None
7	BookingGuests	• Primary keys: BookingGuestId • Foreign keys: BookingId → Bookings(BookingId) • Unique keys: None
8	MenuCategories	• Primary keys: CategoryId • Foreign keys: None • Unique keys: None
9	MenuItems	• Primary keys: MenuItemId • Foreign keys: CategoryId → MenuCategories(CategoryId) • Unique keys: None
10	DiningTables	• Primary keys: TableId • Foreign keys: None • Unique keys: None
11	TableReservations	• Primary keys: ReservationId • Foreign keys: UserId → Users(UserId), TableId → DiningTables(TableId) • Unique keys: None
12	FoodOrders	• Primary keys: FoodOrderId • Foreign keys: BookingId → Bookings(BookingId) • Unique keys: None
13	FoodOrderItems	• Primary keys: OrderItemId • Foreign keys: FoodOrderId → FoodOrders(FoodOrderId), MenuItemId → MenuItems(MenuItemId) • Unique keys: None
14	KOT	• Primary keys: KotId • Foreign keys: FoodOrderId → FoodOrders(FoodOrderId) • Unique keys: FoodOrderId
15	Tours	• Primary keys: TourId • Foreign keys: None • Unique keys: None
16	TourSchedules	• Primary keys: ScheduleId • Foreign keys: TourId → Tours(TourId), GuideId → Users(UserId) • Unique keys: None
17	TourBookings	• Primary keys: TourBookingId • Foreign keys: BookingId → Bookings(BookingId), ScheduleId → TourSchedules(ScheduleId) • Unique keys: None
18	Reviews	• Primary keys: ReviewId • Foreign keys: UserId → Users(UserId)

		• Unique keys: None
19	GuestFolios	• Primary keys: FolioId • Foreign keys: BookingId → Bookings(BookingId) • Unique keys: BookingId
20	FolioTransactions	• Primary keys: TransactionId • Foreign keys: FolioId → GuestFolios(FolioId) • Unique keys: None
21	Invoices	• Primary keys: InvoiceId • Foreign keys: FolioId → GuestFolios(FolioId) • Unique keys: FolioId
22	Payments	• Primary keys: PaymentId • Foreign keys: InvoiceId → Invoices(InvoiceId) • Unique keys: None
23	Refunds	• Primary keys: RefundId • Foreign keys: BookingId → Bookings(BookingId) • Unique keys: None
24	Notifications	• Primary keys: NotificationId • Foreign keys: UserId → Users(UserId) • Unique keys: None

2. Code Packages

2.1 Package Diagram

Figure III.2: Packages Diagram

2.2 Package Descriptions